Bayesian Non-Stationary Spectral Density Estimation from Time–Frequency Data with Tensor-Product P-Splines

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We present a Bayesian framework for estimating smoothly time-varying power spectral densities (PSDs) from time–frequency data. The log-PSD surface $\log S(t, f)$ is modelled by a tensor-product P-spline with an anisotropic roughness prior, sampled in a whitened parametrisation that removes the hierarchical funnel and yields stable Hamiltonian Monte Carlo. The model is independent of how the data are produced: a single Gaussian-coefficient likelihood couples the same surface to different time–frequency transforms, here the Wilson–Daubechies–Meyer (WDM) wavelet transform and the short-time Fourier transform, with the classical moving periodogram recovered as the phase-discarded special case of the latter. On a locally stationary process with known truth, both likelihoods recover the surface with near-nominal credible intervals. Applied to simulated LISA data, the framework recovers the annual modulation of the cyclostationary Galactic confusion foreground, which a stationary noise model — flat at the year-averaged level — cannot represent. Because the likelihood admits a signal mean, the same model extends to joint blocked-Gibbs inference of the noise PSD and an embedded source. The result is a modular framework, applicable to any time–frequency representation whose coefficients admit a local-power likelihood.

I. INTRODUCTION

Classical spectral analysis rests on the assumption that the process under study is stationary. For a stationary series the periodogram ordinates are asymptotically independent and exponentially distributed about the power spectral density (PSD), the structure that underpins the classical Whittle likelihood [1, 2] and the modern Bayesian nonparametric estimators built upon it. These estimators place a smoothing prior on the log-spectrum, for example a penalised spline (P-spline) with a roughness penalty [3], so that a single noisy realisation yields a regularised estimate of $S(f)$ with calibrated uncertainty. We give this stationary likelihood explicitly in Sec. III. It is the starting point that the present work generalises to the non-stationary, time–frequency setting.

Real instruments rarely satisfy the stationarity assumption. Detector noise drifts, glitches, and responds to environmental and instrumental conditions that change on timescales short compared with a typical observation, so that its spectral content evolves in time. Treating such noise as stationary misspecifies the very likelihood on which downstream inference depends, and the resulting bias propagates into any astrophysical parameters estimated against that noise model.

The Laser Interferometer Space Antenna (LISA) is a striking example. Its dominant low-frequency foreground, the unresolved Galactic confusion noise, is non-stationary by construction: as the constellation orbits the Sun, its antenna pattern sweeps across the anisotropic Galactic source distribution, modulating the confusion power with an annual period. This makes the foreground a *cyclostationary* process, well described by a frequency-independent amplitude modulation of a fixed spectral shape [4, 5], and recent stochastic and global-fit pipelines characterise it by iterative spectral fitting [6] or by fitting the confusion amplitude independently in short time segments [7, 8]. The foreground is simultaneously entangled with the resolvable sources one wishes to characterise, so the noise PSD and the source parameters must ultimately be inferred together [9]. In this setting a stationary PSD assumption is not a mild approximation but a structural misspecification, which makes LISA a natural proving ground for non-stationary spectral estimation.

A non-stationary process is most naturally described in a joint time–frequency representation. Two complementary choices are the moving, or short-time, periodogram, which slides a window along the series, and critically-sampled wavelet transforms such as the Wilson–Daubechies–Meyer (WDM) transform [10], which tile the plane into a near-orthogonal grid. The two produce different data products, but both estimate the same underlying object: a PSD that evolves slowly across the time–frequency plane. When that evolution is smooth, recovering it becomes a bivariate regression problem, fitting a smooth surface to noisy local-power observations, to which penalised splines are well suited.

The present work builds directly on the Bayesian nonparametric framework of Tang *et al.* [11], who introduced the dynamic Whittle likelihood for locally stationary processes from the moving periodogram and paired it with a bivariate Bernstein–Dirichlet process prior on the time–frequency spectrum, establishing sup-norm posterior consistency, L_2 contraction rates, and a Bayes-factor test for stationarity. We retain their dynamic Whittle likelihood as one of our observation models, but replace the Bernstein–Dirichlet prior with a tensor-product P-spline. This change buys three things: anisotropic roughness control with separate smoothing in time and frequency, a whitened parametrisation that makes the

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surface amenable to gradient-based (Hamiltonian) sampling, and a Gaussian-coefficient, phase-retaining likelihood that admits a signal mean and hence a joint source–noise fit, which the power-only dynamic Whittle cannot support. We use the reference implementation of Tang *et al.* [11] in the beyondWhittle package as an external baseline in our simulation study (Sec. IV). More broadly, Bayesian estimation of time-varying spectra has a substantial history outside the gravitational-wave setting. **[Avi: Add and cite the adjacent non-stationary spectral literature here, e.g. AdaptSPEC (Rosen, Wood & Stoffer 2012), Dahlhaus locally stationary processes (1997, 2000), and SLEX (Ombao et al.). One or two sentences situating us against the spline-based and segmentation-based alternatives.]**

This paper develops the framework in four steps. Section II introduces the tensor-product P-spline model for the log-PSD surface, with its anisotropic roughness prior and whitened parametrisation for stable HMC. Section III shows that a single Gaussian-coefficient likelihood connects two transforms to that model, the WDM wavelet and short-time Fourier coefficients, with the moving periodogram the phase-discarded special case of the latter. Section V demonstrates it on simulated LISA data carrying a cyclostationary Galactic foreground, which a stationary model cannot represent but both transforms recover. Section VI extends the likelihood to a joint, blocked-Gibbs fit of the noise PSD and an embedded source.

II. BAYESIAN MODEL

We define the statistical model for the time-varying PSD *independently of the data representation*: the surface, prior, and inference machinery below are shared by every observation model in Sec. III.

A. Time-varying PSD model

Let $S(t,f) > 0$ denote the power spectral density at time t and frequency f, assumed to vary smoothly in both arguments. We model its logarithm,

$$\Lambda(t,f) \;=\; \log S(t,f), \tag{1}$$

working on the log scale so that the surface is unconstrained and the multiplicative dynamic range of the PSD becomes additive. The estimand is $\Lambda(t,f)$ on a time–frequency grid (t_n, f_m) supplied by the chosen representation (Sec. III). The model below does not depend on how that grid arises.

B. Tensor-product P-spline representation

We represent the log-PSD surface as a tensor-product P-spline [12, 13],

$$\Lambda(t_i, f_j) = \sum_{k=1}^{K_t} \sum_{l=1}^{K_f} w_{kl}\, B_k^{(t)}(t_i)\, B_l^{(f)}(f_j), \tag{2}$$

where $B_k^{(t)}$ and $B_l^{(f)}$ are B-spline bases of degree d_t, d_f in time and frequency, K_t, K_f are the corresponding basis dimensions, and w_{kl} are the spline coefficients. In matrix form, with $\mathbf{\Lambda} \in \mathbb{R}^{n_t \times n_f}$ the surface on the grid, $\mathbf{B}_t \in \mathbb{R}^{n_t \times K_t}$, $\mathbf{B}_f \in \mathbb{R}^{n_f \times K_f}$ the basis matrices, and $\mathbf{W} \in \mathbb{R}^{K_t \times K_f}$ the coefficient matrix,

$$\mathbf{\Lambda} = \mathbf{B}_t\, \mathbf{W}\, \mathbf{B}_f^{\top}. \tag{3}$$

Equivalently, with $\mathbf{w} = \mathrm{vec}(\mathbf{W})$, $\mathrm{vec}(\mathbf{\Lambda}) = (\mathbf{B}_f \otimes \mathbf{B}_t)\,\mathbf{w}$ [14]. In practice we never form the Kronecker design and evaluate the dense product directly.

Smoothness is imposed by difference penalties in each direction. With $\mathbf{P}_t$ and $\mathbf{P}_f$ the marginal difference-penalty matrices (orders m_t, m_f), the anisotropic penalty on $\mathbf{w}$ has the Kronecker-sum structure [13]

$$\mathbf{Q}(\phi_t, \phi_f) = \phi_t\,(\mathbf{I}_{K_f} \otimes \mathbf{P}_t) + \phi_f\,(\mathbf{P}_f \otimes \mathbf{I}_{K_t}), \text{ [src]} \tag{4}$$

with $\phi_t, \phi_f > 0$ the time and frequency smoothing precisions. The penalty defines a Gaussian roughness prior on the coefficients,

$$\mathbf{w} \mid \phi_t, \phi_f \sim \mathcal{N}\big(\mathbf{0},\, \mathbf{Q}^{-1}\big)\,, \tag{5}$$

so that all smoothing information is carried by the two penalties and the two precisions [3, 14, 15].

Sampling $\mathbf{w}$ jointly with (ϕ_t, ϕ_f) produces the classic hierarchical funnel: the conditional scale of $\mathbf{w}$ collapses as the precisions grow, stalling gradient-based MCMC. **[Avi: cite the funnel pathology (Neal 2003) and the non-centered/whitened reparametrisation remedy (Papaspiliopoulos, Roberts & Sköld 2007).]** We remove it by sampling in the penalty eigenbasis. With $\mathbf{P}_t = \mathbf{U}_t\,\mathrm{diag}(\lambda^t)\,\mathbf{U}_t^{\top}$ and $\mathbf{P}_f = \mathbf{U}_f\,\mathrm{diag}(\lambda^f)\,\mathbf{U}_f^{\top}$, the Kronecker terms are simultaneously diagonalised by $\mathbf{U}_f \otimes \mathbf{U}_t$, so $\mathbf{Q}$ has eigenvalues $d_{ab} = \phi_t \lambda_a^t + \phi_f \lambda_b^f$. We sample standard-normal $s_{ab} \sim \mathcal{N}(0,1)$ and set

$$\mathbf{W} = \mathbf{U}_t\, \mathbf{Z}\, \mathbf{U}_f^{\top}, \qquad Z_{ab} = \frac{s_{ab}}{\sqrt{d_{ab}}}, \text{ [src]} \tag{6}$$

which reproduces $\mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1})$ exactly while making $\mathbf{s}$ *a priori* independent of (ϕ_t, ϕ_f) and absorbing the $|\mathbf{Q}|^{1/2}$ normalisation. The shared penalty null space (bilinear trends in t, f, $d_{ab} = 0$) carries no smoothing information and is given a fixed weak precision τ_0.

Each smoothing precision is given a Gamma hyperprior [3],

$$\phi_t,\ \phi_f \sim \mathrm{Gamma}(\alpha_\phi, \beta_\phi), \tag{7}$$

sampled on the unconstrained log scale, with the Gamma density and its log-Jacobian supplied as a factor. The weakly informative defaults place the prior mass over the range of smoothing scales spanned by the basis, in the spirit of penalised-complexity priors [16, 17].

C. Posterior inference

Let $\mathcal{L}(\mathbf{\Lambda})$ denote the observation-model log-likelihood (Sec. III), which depends on the data only through the surface $\mathbf{\Lambda}$. The joint posterior over the whitened eigen-coefficients and the smoothing precisions is

$$p(\mathbf{s}, \phi_t, \phi_f \mid \text{data}) \;\propto\; \exp\big(\mathcal{L}(\mathbf{\Lambda}(\mathbf{s}, \phi_t, \phi_f))\big)\, p(\mathbf{s})\, p(\phi_t)\, p(\phi_f), \tag{8}$$

with $p(\mathbf{s}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{\Lambda}$ obtained from $(\mathbf{s}, \phi_t, \phi_f)$ through the deterministic whitening map of Eq. (6). No explicit penalty factor appears. We draw samples with the No-U-Turn Sampler [NUTS, 18], using standard step-size and mass-matrix adaptation during warm-up. Each chain is initialised from a penalised least-squares fit of $\mathbf{\Lambda}$ to the log of the local-power data, mapped into the whitened coordinates, and both observation models use the same warm-start procedure on their respective local-power statistics.

A single likelihood evaluation is then cheap. Given the current smoothing precisions ϕ_t, ϕ_f (sampled on the log scale, with $\phi_t, \phi_f \sim \text{Gamma}(\alpha_\phi, \beta_\phi)$) and the whitened eigen-coefficients $s_{ab} \sim \mathcal{N}(0, 1)$, we form the eigenvalues $d_{ab} = \phi_t \lambda_a^t + \phi_f \lambda_b^f$ (with the null space held at the fixed precision τ_0), set $Z_{ab} = s_{ab}/\sqrt{d_{ab}}$, assemble the log-spline surface $\mathbf{\Lambda} = (\mathbf{B}_t \mathbf{U}_t)\, \mathbf{Z}\, (\mathbf{B}_f \mathbf{U}_f)^\top$, and evaluate the observation-model log-likelihood $\mathcal{L}(\mathbf{\Lambda})$ of Sec. III. The basis and eigenvector products are precomputed once, so each evaluation is a pair of dense matrix products followed by the likelihood.

Implementation. We implement the model in JAX [19] with NumPyro [20], exploiting automatic differentiation and just-in-time (JIT) compilation for gradient-based sampling. The smoothing precisions ϕ_t, ϕ_f are sampled on the log scale to keep the sampler away from the boundary at zero, and the time and frequency grids are standardised to $[0, 1]$ for numerical stability of the B-spline basis evaluation. Time knots are placed adaptively in the spirit of Maturana-Russel and Meyer [3], clustering where the spectrum varies fastest. The shared model and sampler configuration is collected in Table I. The per-study warm-up and draw counts, which differ between the noise-only and joint fits, are given with each study below. A single one-year joint A/E fit (two channels, signal and non-stationary noise) completes in $\simeq 2.5$ min of wall time on a single workstation CPU core with no GPU. **[Avi: Confirm the hardware (CPU model / core count) for the wall-clock figure.]**

Convergence diagnostics. We assess convergence with standard chain diagnostics [21, 22]: the rank-normalised Gelman–Rubin statistic $\hat{R} < 1.01$, bulk and tail effective sample sizes above a few hundred, and the absence of divergent transitions and maximum tree-depth saturation. No fit in any study produced sampler divergences. **[Avi: Report the actual $\hat{R}$/ESS for a representative fit — these are not currently logged. Add logging to the study scripts.]**

TABLE I. Default model and sampler configuration. Interior-knot counts set the basis dimension as $K = (\text{interior knots}) + d + 1$. The moving-periodogram fits use 6 frequency knots in place of 10. Warm-up/draw counts vary by study and are stated in the text: 250/250 for the simulation study, 300/250 for the LISA noise ensemble, and 800/2000 for the time-localised-source coverage study.

Quantity	Symbol	Value
B-spline degree (time, freq)	d_t, d_f	3, 3
Interior knots (time, freq)	–	8, 10
Difference-penalty order	m_t, m_f	2, 2
Smoothing hyperprior	$(\alpha_\phi, \beta_\phi)$	(2, 1)
Null-space precision	τ_0	10^{-4}
NUTS target accept probability	–	0.85
NUTS max tree depth	–	10

III. OBSERVATION MODELS

The model of Sec. II specifies a prior over the surface $\mathbf{\Lambda} = \log \mathbf{S}$. A likelihood linking a time–frequency data product to it completes the posterior. We give a likelihood for each of two transforms, the moving (short-time Fourier) periodogram and the WDM wavelet transform, and show in Sec. III C that both are special cases of a single Gaussian-coefficient form, with the moving periodogram the phase-discarded version of the Fourier likelihood. Only $\mathcal{L}(\mathbf{\Lambda})$ differs — the PSD model is unchanged. The two front ends sample the plane very differently (Fig. 1): the WDM transform tiles it into a regular, near-orthogonal grid, whereas the moving periodogram returns a scattered sawtooth that does not fill it.

The stationary case fixes the reference point. When the PSD is constant in time, $S(t, f) = S(f)$, the periodogram ordinates are asymptotically independent and exponentially distributed about $S(f)$, giving the classical Whittle likelihood [1, 2]

$$L_{\mathrm{W}}(S) = \prod_k \frac{1}{S(f_k)} \exp\left(-\frac{I(f_k)}{S(f_k)}\right), \tag{9}$$

with $I(f_k)$ the periodogram at Fourier frequency f_k. Each time–frequency likelihood below reduces to this form within a single short window or tile, so Eq. (9) is the stationary limit that the non-stationary models generalise.

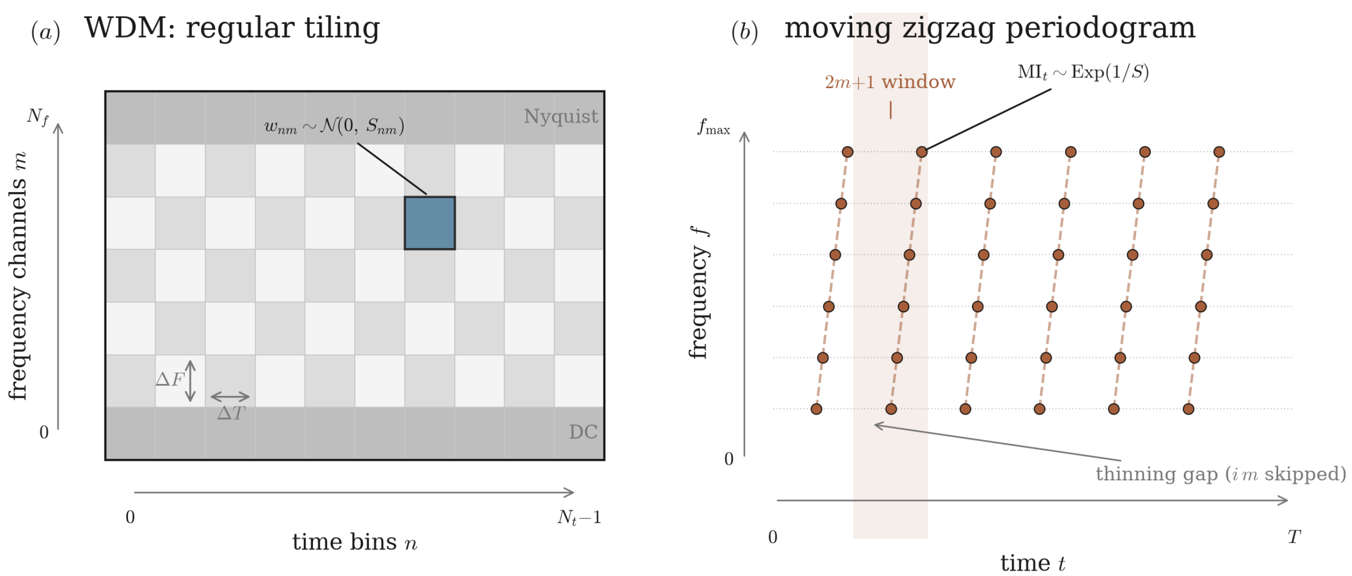

FIG. 1. The two time–frequency front ends that feed the same P-spline surface. *(a)* The WDM transform tiles the plane into a regular, near-orthogonal grid, contributing one χ_1^2 coefficient $w_{nm} \sim \mathcal{N}(0, S_{nm})$ per cell of size $\Delta T \times \Delta F$. *(b)* The thinned zigzag moving periodogram of Tang *et al.* [11] does not fill the plane: the evaluated Fourier frequency cycles with the time index, tracing rising diagonal ramps, and the thinning skips $i\,m$ ordinates between blocks, leaving a scattered sawtooth of χ_2^2 ordinates $\mathrm{MI}_t \sim \mathrm{Exp}(1/S)$, each formed from its own overlapping $2m+1$ window. The same whitened P-spline surface of Sec. II is fitted to either sampling.

A. Moving periodogram likelihood

We slide a short window of length $2m+1$ along the series and form a local periodogram at each step [11]. The moving periodogram ordinates MI_t of order m, evaluated at time point $t = 1, \ldots, T$ from the samples $X_{t-m+1}, \ldots, X_{t+m}$, are

$$\mathrm{MI}_t = \frac{1}{2\pi(2m+1)} \left| \sum_{\nu=0}^{2m} X_{\nu+t-m} \exp\bigl(-i\pi\nu\, \lambda_{\mathrm{mod}(t)}\bigr) \right|^2 , \tag{10}$$ [src]

where $\lambda_j = \frac{2j}{2m+1}$, $j = 1, \ldots, m$, are the Fourier frequencies and $\mathrm{mod}(t) = 1 + ((t-1) \bmod m)$.

For a locally stationary process these ordinates are approximately independent and exponentially distributed with mean S, which gives the dynamic Whittle likelihood

$$L_{\mathrm{DW}}^{(1)}(S) = \prod_{t=1}^{T} \frac{1}{S(t/T, \lambda_{\mathrm{mod}(t)})} \exp\left(-\frac{\mathrm{MI}_t}{S(t/T, \lambda_{\mathrm{mod}(t)})} \right) . \tag{11}$$

The overlapping windows induce dependence between neighbouring ordinates, so we follow Tang *et al.* [11] and use a thinned variant that retains a zigzag subset. With thinning factor i,

$$L_{\mathrm{DW}}^{(i)}(S) = \prod_{l=1}^{B_i} \prod_{j=1}^{m} \frac{1}{S(u_{j,l,i}, \lambda_j)} \exp\left(-\frac{\mathrm{MI}(u_{j,l,i}, \lambda_j)}{S(u_{j,l,i}, \lambda_j)} \right) , \tag{12}$$ [src]

where $B_i = \lfloor (T-m)/(im) \rfloor$ and $u_{j,l,i} = (i(l-1)m+j)/T$.

In the notation of Sec. II, $\mathcal{L}(\boldsymbol{\Lambda}) = \log L_{\mathrm{DW}}^{(i)}$ with $S = e^{\Lambda}$ evaluated at the ordinate locations. The dynamic Whittle treats the thinned ordinates as independent and exponential, which is exact only asymptotically and requires the spectrum to be approximately constant across a window. Thinning trades data for reduced inter-ordinate correlation, and because it works with the power periodogram it discards phase.

B. WDM likelihood

The Wilson–Daubechies–Meyer (WDM) transform [10] tiles the data into a near-orthogonal time–frequency grid of coefficients w_{nm}, where n indexes WDM time bins and m frequency channels. For a locally stationary process the diagonal approximation models each pixel as a zero-mean Gaussian whose variance is the local evolutionary power,

$$w_{nm} \sim \mathcal{N}(0,\, S_{nm}) , \qquad S_{nm} = S(t_n, f_m). \tag{13}$$

The resulting log-likelihood, with $\boldsymbol{\Lambda} = \log \mathbf{S}$, is

$$\mathcal{L}(\boldsymbol{\Lambda}) = -\frac{1}{2} \sum_{n,m} \left[\log(2\pi) + \Lambda_{nm} + w_{nm}^2\, e^{-\Lambda_{nm}} \right] . \tag{14}$$ [src]

Each pixel contributes a single χ_1^2 observation. The smoothing prior of Sec. II B supplies the regularisation that makes single-realisation estimation feasible. The likelihood treats the WDM coefficients as independent, which is exact only for a perfectly orthogonal transform — residual nearest-neighbour time correlation remains, and as with the moving periodogram the spectrum is assumed approximately constant within a tile.

C. One model, two transforms

Both likelihoods are instances of a single Gaussian-coefficient form. A time–frequency cell with R real components contributes

$$-\tfrac{1}{2}\sum_{n,m}\Big[R\,\Lambda_{nm}+\big(\textstyle\sum_{r=1}^{R} c_{nm}^{(r)2}\big)\,e^{-\Lambda_{nm}}\Big], \text{[src]} \tag{15}$$

so the WDM coefficient ($R=1$, χ^2_1) and the complex short-time Fourier coefficient ($R=2$, real and imaginary parts, χ^2_2, whose squared modulus is the periodogram) are handled by the *same* likelihood. The two transforms that enter the framework are therefore the WDM wavelet transform and the short-time Fourier transform (STFT), each fitted with this Gaussian-coefficient likelihood.

The moving periodogram is not a third transform but a special case of the Fourier likelihood: the exponential dynamic-Whittle form of Eq. (11) is exactly the $R=2$ marginal of Eq. (15) over the coefficient phase. The dynamic Whittle thus models the Fourier power, whereas the full $R=2$ form keeps the phase. Retaining phase matters because it lets the coefficient carry a signal mean, $c\sim\mathcal{N}(h(\boldsymbol{\theta}),S)$, which is what enables the joint source–noise inference of Sec. V. The power-only dynamic Whittle cannot. In all cases the time series is mapped to (time grid, frequency grid, coefficients) by one of the two transforms, and the identical whitened P-spline model of Sec. II fits the surface.

IV. SIMULATION STUDY

We assess statistical performance where the truth is known, fitting the P-spline model through the WDM coefficient likelihood and the moving-periodogram dynamic Whittle and comparing each against the true surface. The phase-retaining Fourier likelihood adds nothing in this noise-only test — without a signal mean it reduces to the same power statistic — so we defer it to the LISA demonstration (Sec. V).

A. Locally stationary process

Following Tang *et al.* [11], we use the locally stationary time-varying MA(1) process (LS2) with i.i.d. standard-normal innovations $\{w_t\}$,

$$X_{t,T}=w_t+1.1\cos\!\big(1.5-\cos\!\big(4\pi\tfrac{t}{T}\big)\big)\,w_{t-1},\qquad t=1,\dots,T, \tag{16}$$

with analytic pointwise PSD $f_0(u,\omega)=1+b_1(u)^2+2b_1(u)\cos\omega$, $b_1(u)=1.1\cos(1.5-\cos 4\pi u)$. For each realisation we fit the model through the WDM likelihood, Eq. (14), and through the moving-periodogram likelihood, Eq. (12), and repeat across four observation lengths $n=n_t n_f$ with $n_f=24$ frequency bins and $n_t\in\{24,48,96,192\}$ time bins, i.e. $n\in\{576,1152,2304,4608\}$. Each fit draws 250 warm-up and 250 posterior samples with a single NUTS chain under the configuration of Table I.

The WDM estimator targets the expected local power $\mathbb{E}[w_{nm}^2]$, which equals the digital PSD up to a near-constant per-channel normalisation, $\mathbb{E}[w_{nm}^2]=C_m f_0(t_n,f_m)$. We calibrate C_m once from unit white noise and report the error against both the Monte Carlo $\mathbb{E}[w^2]$ target and the calibrated analytic PSD $C_m f_0$.

B. Metrics

We summarise point-estimate accuracy with the mean squared error of the log-PSD,

$$\mathrm{MSE}_{\log f}=\frac{1}{T(K+1)}\sum_{t=1}^{T}\sum_{j=0}^{K}\Big(\ln\hat f\big(\tfrac{t}{T},\tfrac{j}{K}\big)-\ln f_0\big(\tfrac{t}{T},\tfrac{j}{K}\big)\Big)^2, \text{[src]} \tag{17}$$

using the posterior mean as the point estimate. We also report the empirical coverage of the posterior 90% credible intervals and, as a measure of posterior contraction, the width of the 90% credible interval on $\log S$ averaged over the grid (a scale-free quantity, comparable across representations). Each configuration is summarised over 100 independent realisations. We plot the median and interquartile range across realisations.

As an external baseline we run the Bernstein–Dirichlet dynamic-Whittle estimator of Tang *et al.* [11] on the same realisations through its beyondWhittle reference implementation. Because that estimator and our moving-periodogram fit share the dynamic Whittle likelihood and differ only in the prior (Bernstein–Dirichlet process versus tensor-product P-spline), the comparison isolates the effect of the prior under matched data. **[Avi: Run beyondWhittle on the stored LS2 realisations and add its MSE/coverage/CI-width curves to Fig. 3.]**

C. Results

Both observation models recover the time-varying surface and contract with the data (Fig. 3). Over 100 realisations at each of four observation lengths ($n=576$ to 4608), the median $\mathrm{MSE}_{\log f}$ falls monotonically with n for both likelihoods, the 90% credible-interval coverage stays close to nominal, and the credible-interval width on $\log S$ contracts steadily with n, as expected. No sampler divergence occurred in any fit. On this smooth process the moving periodogram has the lower median $\mathrm{MSE}_{\log f}$ at matched n, though the gap is comparable to the interquartile spread across realisations **[Avi: state**

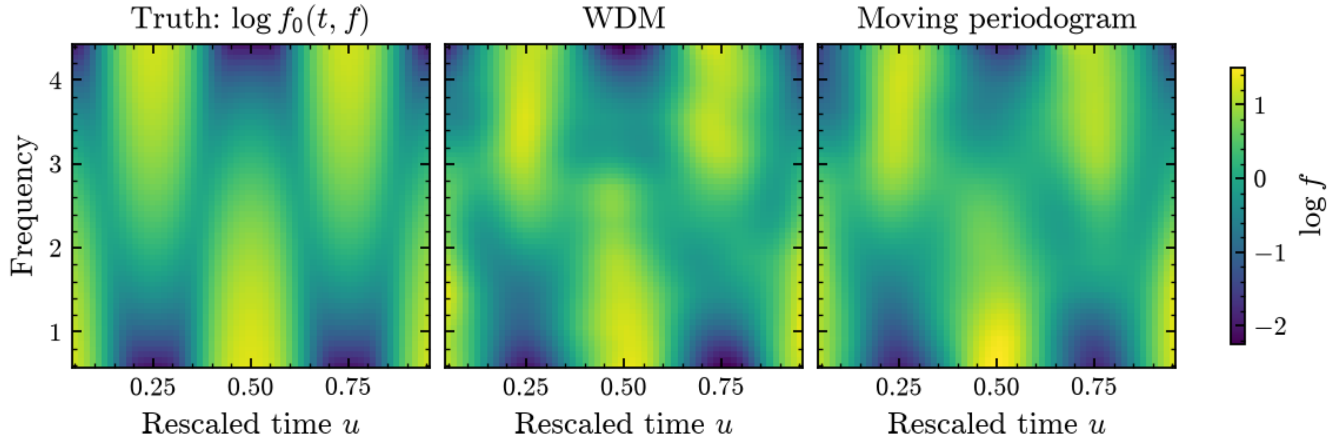

FIG. 2. Single LS2 realisation. Left: true time-varying PSD $\log f_0(t, f)$. Middle: posterior-median $\log \hat{S}(t, f)$ from the WDM likelihood. Right: posterior-median $\log \hat{S}(t, f)$ from the moving-periodogram likelihood. All panels share a colour scale and are rescaled to the analytic PSD scale.

whether the separation exceeds the IQR band (i.e. whether it is significant). If not, soften to "comparable"]. We return in Sec. VII to why the WDM front end is nonetheless preferable for phase-coherent inference.

V. RECOVERING A NON-STATIONARY LISA FOREGROUND

We now apply the framework to simulated LISA data, where the non-stationarity is physical rather than imposed. As described in Sec. I, the Galactic confusion foreground is cyclostationary, its power modulated with an annual period [4]. We show that both transforms of Sec. III recover this time-varying noise PSD, which a stationary model cannot represent.

A. Simulated data

We simulate one year of a second-generation TDI channel (TDI-X) **[Avi: cite TDI (Tinto & Dhurandhar 2021 review, Armstrong, Estabrook & Tinto 1999) and the LISA mission (Amaro-Seoane et al. 2017).]** sampled at interval $\Delta t \simeq 200\,\mathrm{s}$ ($N \simeq 1.6 \times 10^5$ samples, a Nyquist frequency of $2.5\,\mathrm{mHz}$), in fractional-frequency units. The noise has two independent components,

$$n(t) = n_{\mathrm{inst}}(t) + \sqrt{r(u)}\, n_{\mathrm{gal}}(t), \tag{18}$$

with $u = t/T_{\mathrm{obs}} \in [0, 1]$ rescaled time. The instrument component n_{inst} and the Galactic-confusion component n_{gal} are drawn as independent zero-mean coloured Gaussian processes whose one-sided PSDs are the analytic Robson–Cornish–Liu fits [23] $S_{\mathrm{inst}}(f)$ and $S_{\mathrm{gal}}(f)$: each is generated in the frequency domain with $\tilde{n}(f_k) \sim \mathcal{CN}\big(0,\, S(f_k)/(4\,\Delta f\,\Delta t^2)\big)$ and transformed to the time domain, so the realised one-sided PSD reproduces the target.

The confusion power is modulated by the cyclostationary annual law of Digman and Cornish [4],

$$r(u) = 1 + \sum_{k=1}^{5} A_k \cos\big(2\pi k\, u\, T_{\mathrm{obs}}/T_{\mathrm{yr}} - \varphi_k\big), \tag{19}$$

with the tabulated harmonic coefficients (A_k, φ_k) of their Table 1 (A-channel, one-year fit). The second harmonic dominates, so the confusion power peaks twice per year and varies by a factor $\simeq 7$, reproducing their Figure 1. The coefficients average to $\langle r \rangle = 1$ over whole years, so the modulation redistributes confusion power in time without changing its time average. Because the two components are independent and r varies slowly relative to a WDM time bin, the instantaneous one-sided PSD is

$$S(u, f) = S_{\mathrm{inst}}(f) + r(u)\, S_{\mathrm{gal}}(f), \tag{20}$$

which is exactly the local power the estimator targets. The estimator recovers the total surface $S(u, f)$, not the individual components. Eq. (20) is a property of the simulated data, not an output of the fit. This is the discriminating regime: a stationary model fits only a single time-averaged spectrum, whereas $S(u, f)$ carries the annual modulation. The data also contain a resolvable Galactic binary at $f_0 = 1.5\,\mathrm{mHz}$ (optimal SNR 200, generated with jaxGB in the same TDI-X generation-2 convention) **[Avi: cite jaxGB and the fast Galactic-binary waveform it implements (Cornish & Littenberg 2007)]**. Recovering its parameters jointly with the noise is the subject of Sec. VI. For the noise-recovery results of this section the binary has been removed, so the estimand is the pure non-stationary noise PSD.

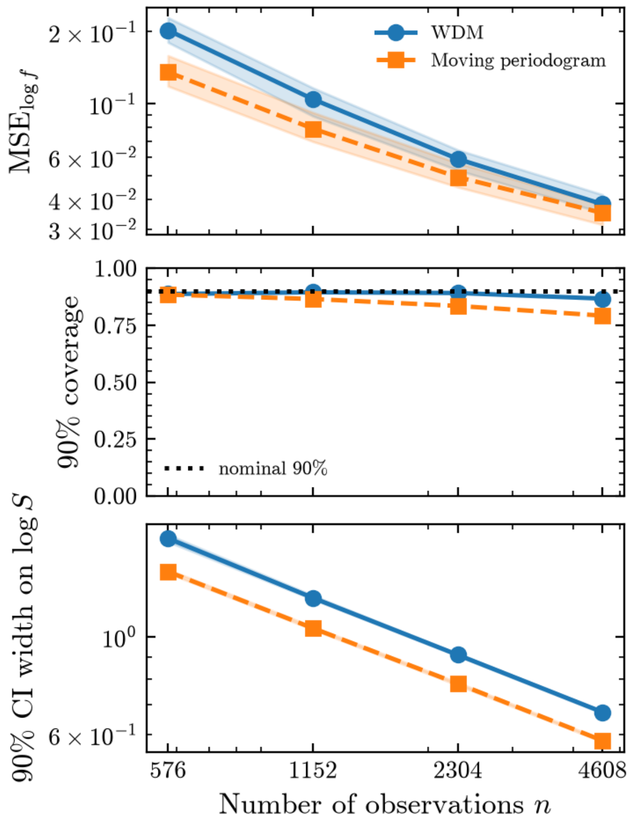

FIG. 3. Performance versus the number of observations n, over 100 realisations per point (median, with interquartile band where shown). Top: $\mathrm{MSE}_{\log f}$. Middle: 90% credible-interval coverage (nominal level dotted). Bottom: median width of the 90% credible interval on $\log S$, contracting with n. One curve per observation model (WDM and moving periodogram).

B. Recovered time-varying PSD

From a single noisy realisation, the estimator denoises the raw χ_1^2 WDM power into a smooth log-PSD surface that matches the Monte Carlo reference (Fig. 4). The recovered $\log \hat{S}(u, f)$ captures both the steep instrument spectrum at low frequency and the twice-yearly Galactic confusion modulation that the raw power buries in noise.

The effect of the stationarity assumption is sharpest in a single frequency channel. Figure 5 shows the recovered noise PSD at $f = 1.5\,\mathrm{mHz}$ over the year: a stationary model is necessarily flat in time at the time-averaged level, overestimating the noise when the Galactic confusion is low and underestimating it when the confusion peaks, by up to the factor-$\simeq 7$ swing of the modulation, whereas the time-varying fit tracks the twice-yearly modulation and contains the true $S(u, f)$ within its 90% band.

The recovery does not depend on the time–frequency representation. We analyse the same non-stationary noise with both transforms of Sec. III: the WDM wavelet coefficients and the short-time Fourier coefficients, the latter in both its phase-retaining form and its power-only dynamic-Whittle reduction. All of them recover the same cyclostationary modulation of the noise PSD in the binary's channel, tracking the true annual envelope where a stationary model is flat (Fig. 6). Because they share the identical P-spline model, the quantity being tested is the stationarity assumption rather than the representation. Over an ensemble of independent realisations the 90% credible band on the recovered noise PSD attains empirical coverage of 0.78 and 0.84 in the two noise-orthogonal A and E channels (Sec. VI), below the nominal 0.90. We attribute this under-coverage to the diagonal likelihood approximation: treating the WDM coefficients as independent neglects the residual nearest-neighbour correlations of a finitely-orthogonal transform (Sec. III B), which narrows the posterior band relative to the true sampling variability. **[Avi: we get 0.78 vs 0.90.. probably because few repeated sims?]**

VI. JOINT SIGNAL–NOISE INFERENCE WITH A BLOCKED-GIBBS SCHEME

[Avi: Work in progress... I want to show that (1) with stationary noise, using $S(f)$ or $S(t, f)$ leads to no biases in physical signal params, (2) with non-stationary noise, the time-varying fits are necessary to avoid biases. Right now, i see little differences...]

The same Gaussian-coefficient likelihood that recovers the noise PSD admits a signal mean, $c_{nm} \sim \mathcal{N}(h_{nm}(\boldsymbol{\theta}), S_{nm})$, so in principle the source parameters $\boldsymbol{\theta}$ and the non-stationary noise surface $\boldsymbol{\Lambda}$ can be inferred *jointly*. We develop a blocked-Gibbs scheme for this joint fit and report preliminary results on a Galactic binary (GB)/ massive-black-hole-binary (MBHB) / Stellar-mass black-hole binary (BBH).

[Avi: Renate has ideas on how to do this for the moving-periodogram dynamic Whittle... But for now lets focus on the WDM.]

A. Blocked Gibbs scheme

Rather than sampling the full state with a single Hamiltonian trajectory, we use a Metropolis-within-Gibbs scheme that alternates two blocks. The PSD block, conditional on the current source estimate, subtracts the signal coefficients $h_{nm}(\boldsymbol{\theta})$ and updates the whitened noise parameters $(\mathbf{s}, \phi_t, \phi_f)$ under the model of Sec. II. The source block, conditional on the current noise surface $\boldsymbol{\Lambda}$, updates the source parameters $\boldsymbol{\theta}$. Blocking is deliberate: the high-dimensional smooth PSD coefficients and the low-dimensional source parameters have very different geometries, so each block's sub-kernel can adapt independently while the Gibbs sweep couples them. The scheme accommodates heterogeneous sub-kernels: a gradient-

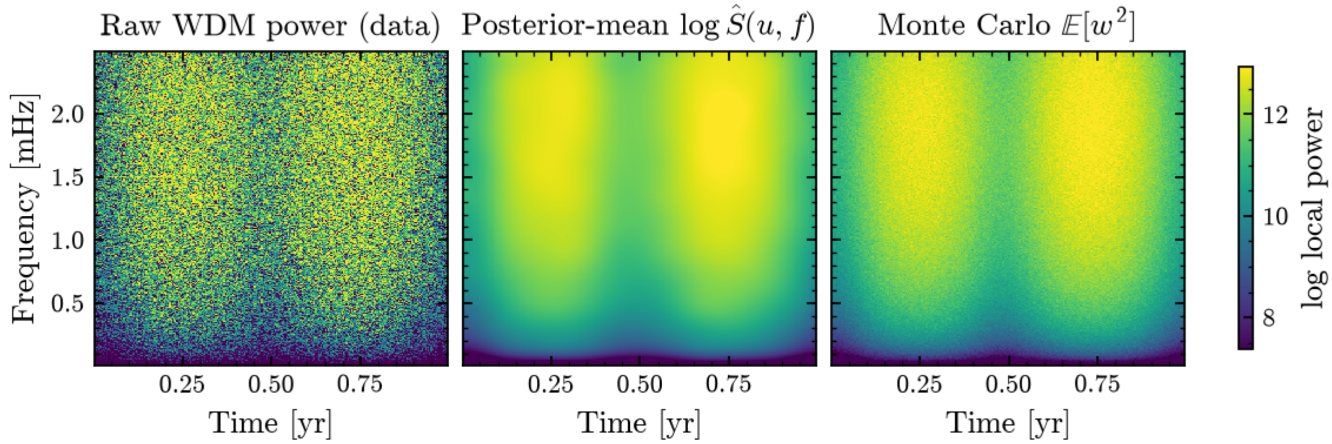

FIG. 4. One year of non-stationary LISA noise, single realisation: raw WDM log power (the data), the posterior-mean $\log \hat{S}(u,f)$, and the Monte Carlo $\mathbb{E}[w^2]$ reference. The estimator denoises the data into the smooth time-varying surface, recovering the twice-yearly Galactic-confusion modulation invisible in the raw power.

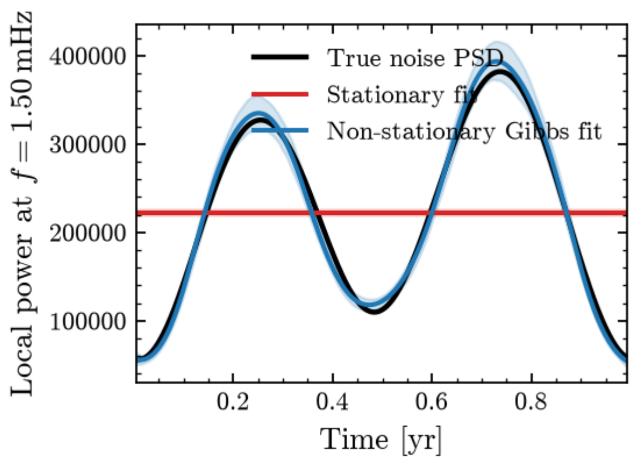

FIG. 5. Recovered TDI noise PSD in the binary's channel ($f = 1.5$ mHz) over one year. A stationary fit (red, with 90% band) is necessarily flat at the time-average and so misses the modulation, while the time-varying fit (blue) tracks the true cyclostationary noise PSD (black), which peaks twice per year.

based NUTS update for the differentiable PSD block, and a gradient-free update for a source model whose waveform is not differentiable. [src]

B. Proof of concept: a Galactic binary

We first demonstrate the joint fit on the resolvable Galactic binary embedded in the data of Sec. V, with both blocks updated by NUTS. From the single one-year realisation the fit recovers the coherent binary together with the noise surface: although the binary is sub-threshold in any single time–frequency cell, its amplitude is recovered to within $\sim 10\%$ by coherent summation across the plane ($|A|/|A|_{\rm true} = 0.90$). As a check that the wavelet domain costs no information relative to the usual analysis, we compare the WDM source posterior in stationary noise against the standard frequency-domain Whittle analysis on the same data and find the two agree across all four binary parameters (Fig. 7).

Repeating the joint fit over the two noise-orthogonal TDI channels A and E, each with its own Digman–Cornish modulation and one shared binary amplitude (each fit 300 warm-up and 250 posterior draws, WDM grid $n_t = 256$ time bins), gives unbiased amplitude recovery across 12 independent one-year realisations, $|A|/|A|_{\rm true} = 1.01 \pm 0.04$ (Fig. 8), with the 90% PSD credible band attaining the coverage quoted in Sec. V B. No realisation produced sampler divergences. **[Avi: 12 realisations is a small ensemble for a coverage statement. Report the binomial Monte Carlo error on the $0.78/0.84$ coverages, or grow the ensemble.]**

C. Why the noise model matters for the source

A year-long binary's parameters are robust to the noise model: the strain is linear in the quadrature amplitudes, so the weighted least-squares estimate is unbiased even when the noise weights are misspecified, and a full-cycle source averages over the modulation. The interesting regime is a *time-localised* source, such as a burst or the final days of an MBHB inspiral, which accumulates its signal-to-noise at a single epoch and is measured against the *instantaneous* noise. We model such a source through its frequency-channel STFT coefficient, $c_n = A\, e^{i(\phi_0 + 2\pi\, \delta f\, t_n)}\, W(u_n) + \text{noise}_n$, with W a localised envelope and noise_n of local variance $S(u_n, f_0)$, and infer $(\delta f, A, \phi_0)$ with NUTS (800 warm-up, 2000 draws) under two noise models: the stationary frequency-domain

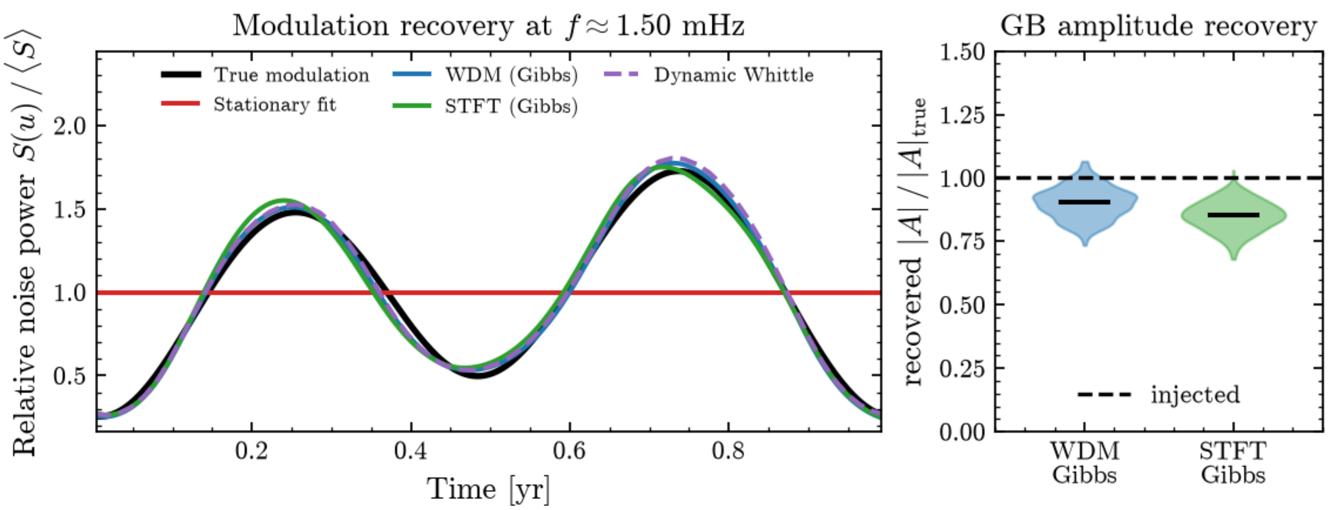

FIG. 6. Non-stationary noise across both transforms. Left: the relative noise-power modulation $S(u)/\langle S\rangle$ in the binary's channel. The WDM (blue), short-time Fourier (green), and dynamic-Whittle moving-periodogram (purple, dashed) fits all track the true cyclostationary envelope (black), while a stationary model is flat (red). Right: posterior of the embedded binary amplitude relative to truth for the two phase-retaining fits (violins with median bars, injected value dashed), a preview of the joint fit of Sec. VI. The dynamic Whittle is power-only and so estimates the noise PSD on the signal-subtracted series.

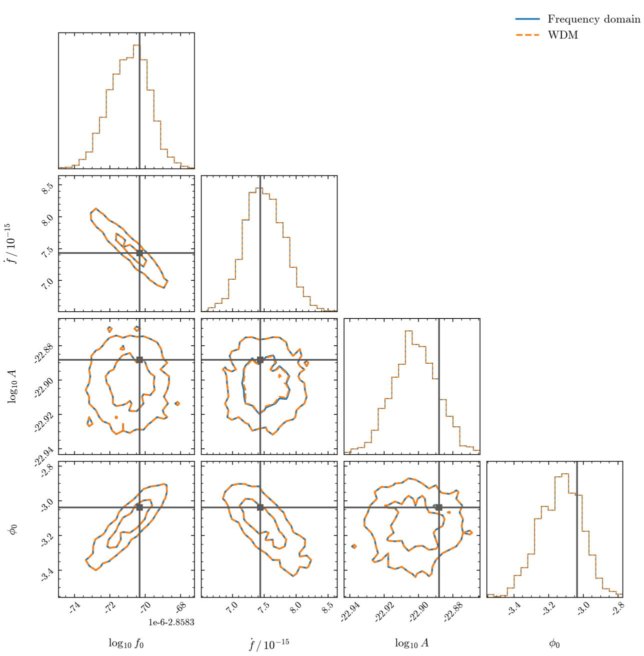

FIG. 7. Galactic-binary parameter posteriors in stationary noise, recovered in the WDM domain (orange, dashed) and with the standard frequency-domain Whittle analysis (blue, solid), injected values in black. The two posteriors coincide across $(\log_{10} f_0, \dot f, \log_{10} A, \phi_0)$, confirming that the wavelet transform sacrifices no information on the source.

Whittle (a single, time-averaged PSD) and the WDM P-spline time-varying PSD.

The two analyses differ markedly (Fig. 9). For a transient at a confusion *minimum*, a quiet epoch, the time-varying PSD recognises the low local noise and returns a posterior a factor $\simeq 2$ tighter than the stationary Whittle in every parameter, both centred on the truth. At a confusion *maximum* the stationary model instead underestimates the local noise and becomes overconfident, its 90% credible region containing the truth only 0.77 of the time against 0.88 for the time-varying model (over 300 realisations, nominal 0.90, binomial MC error ≈ 0.02). Modelling the non-stationarity is therefore what keeps the source posterior both calibrated and maximally informative for time-localised sources – the regime the MBHB demonstration below targets.

VII. DISCUSSION

The framework separates the statistical model, the smooth log-PSD surface and its prior, from the observation model that relates a time–frequency data product to that surface. Section III exercised this separation with two transforms, the WDM wavelet transform and the short-time Fourier transform (the latter used either with phase or, as the moving periodogram, without), both of which share the model of Sec. II verbatim. Any other near-orthogonal representation whose coefficients admit a local-power (Gaussian-coefficient or exponential) likelihood, such as a multitaper spectrogram, a Q-transform, or another wavelet family, can be added without changing the model. **[Avi: Cite multitaper (Thomson 1982) and the Q-transform (Chatterji et al. 2004) if we keep naming them as concrete examples.]**

The LISA demonstration shows that the choice of stationarity assumption, not the choice of representation,

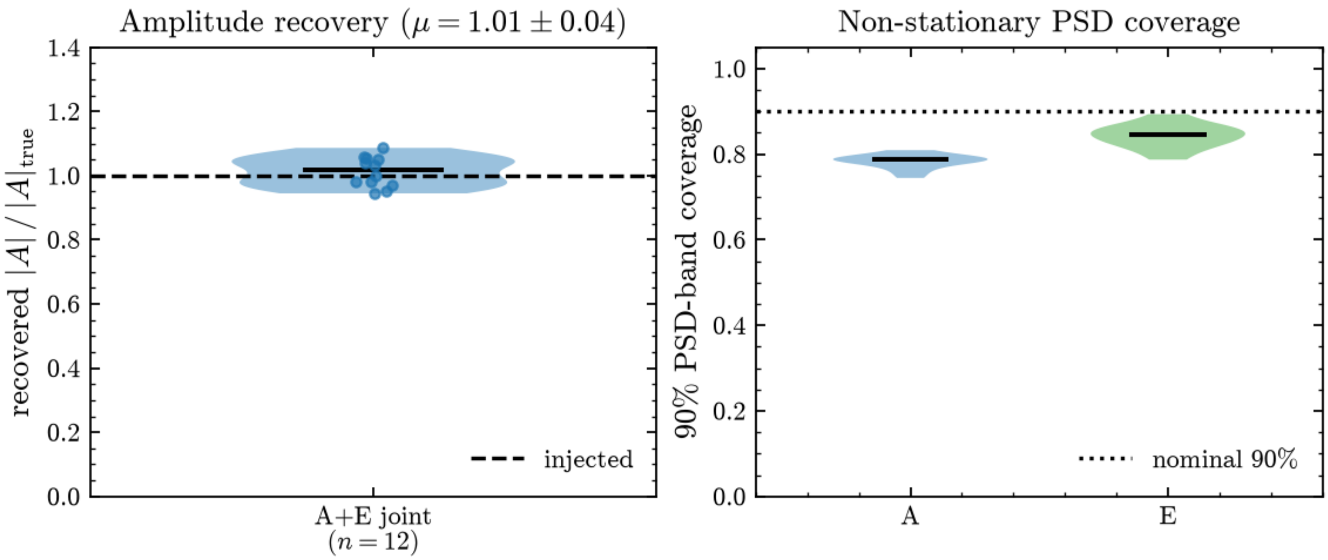

FIG. 8. A/E ensemble over 12 independent one-year realisations of the joint fit. Left: recovered binary amplitude relative to truth (violin with points as individual realisations, median bar, injected value dashed), unbiased at 1.01 ± 0.04. Right: empirical coverage of the 90% PSD credible band per channel against the analytic truth (nominal level dotted), near nominal.

is what drives the bias: the wavelet and Fourier transforms agree with one another and with the frequency-domain analysis where stationarity holds, and both track the cyclostationary modulation where it does not. Existing LISA stochastic pipelines model this modulation either with an analytic cyclostationary template [4, 5, 8], a weakly-parametric spectral model [24], or a per-segment confusion amplitude [7]. Our estimator instead infers the time–frequency PSD surface non-parametrically, requiring no template for the modulation shape. This makes the framework a natural component of detector-characterisation and global-fit pipelines [9], where a misspecified stationary noise model otherwise contaminates the recovered source population.

Several extensions follow naturally. The source block can be generalised from the two linear quadrature amplitudes at fixed frequency to the full parameter vector $\boldsymbol{\theta} = (f_0, \dot{f}_0, A, \ldots)$ with a nonlinear, multimodal frequency search, leaving the blocked structure unchanged. Beyond a single source, the same coefficient likelihood supports multi-source and multi-channel joint fits, and relaxing the diagonal approximation to a correlated-coefficient likelihood would account for the residual nearest-neighbour correlations left by a finitely orthogonal transform. Finally, while a year-long binary's parameters are robust to the noise model, time-localised sources are not (Sec. VI C): there the time-varying PSD is what keeps the source posterior both calibrated and maximally informative. The concrete next step we are pursuing is the massive-black-hole-binary demonstration of Sec. **??**: a full transient waveform and LISA response inside the blocked-Gibbs loop. This is the sharply time-localised regime where we expect critically-sampled wavelets to also outperform the moving periodogram.

VIII. CONCLUSION

We have presented a Bayesian framework for non-stationary spectral estimation in which a single statistical object, a smooth log-PSD surface modelled by a whitened tensor-product P-spline, is decoupled from the time–frequency representation used to observe it. A single Gaussian-coefficient likelihood connects two transforms to that one model, the WDM wavelet transform and the short-time Fourier transform, with the classical moving periodogram recovered as the phase-discarded special case of the latter. On a locally stationary process with known truth the estimator contracts with the data at near-nominal coverage. On simulated LISA data it recovers the cyclostationary Galactic foreground, which a stationary model cannot represent because it is flat at the year-averaged level and misses the full annual swing of the confusion. Because the same coefficient likelihood admits a signal mean, the noise PSD and an embedded source can in principle be inferred jointly with a blocked-Gibbs scheme. We report preliminary results on a Galactic binary and outline the massive-black-hole-binary demonstration that remains. Since any near-orthogonal representation with a local-power likelihood plugs into the same model, the framework is a portable component for detector characterisation and global-fit pipelines wherever the noise drifts in time.

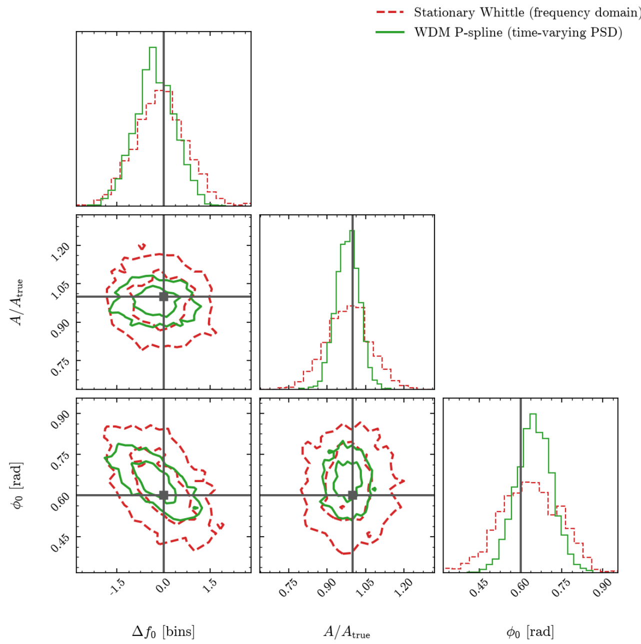

FIG. 9. Parameter posteriors for a time-localised source at a quiet (confusion-minimum) epoch, inferred with the stationary frequency-domain Whittle (red, dashed) and the WDM P-spline time-varying PSD (green, solid), with truth in grey. By weighting the source against the low local noise, the time-varying PSD yields a posterior a factor $\simeq 2$ tighter in every parameter. (At a confusion maximum the roles reverse and the stationary model is instead overconfident, with 90% coverage 0.77 versus 0.88.)

DATA AND CODE AVAILABILITY

[Avi: Add a one-line statement pointing to the public repository and the archived release/DOI used to produce the figures, and name the key dependencies (jaxGB, beyondWhittle, the WDM transform implementation). Note: `ramos2023scikit` (scikit-fda) is in `refs.bib` but currently uncited — cite it where the B-spline basis is built, or drop it.]

ACKNOWLEDGMENTS

[Avi: TODO: funding, collaborators, computing resources.]

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